@import url('https://fonts.googleapis.com/css2?family=Roboto+Flex:opsz,wdth,wght,GRAD@8..144,25..151,100..1000,-200..150&display=swap'); @page { size: A4; margin: 20mm; @bottom-right { content: counter(page); font-family: 'Roboto Flex', sans-serif; font-size: 9px; color: #757688; } } body { font-family: 'Roboto Flex', sans-serif; color: #1a1b24; background-color: #ffffff; line-height: 1.6; font-size: 14px; margin: 0; padding: 0; } h1, h2, h3, h4, h5, h6 { color: #213be7; font-family: 'Roboto Flex', sans-serif; font-weight: 500; margin-top: 1.5em; margin-bottom: 0.5em; page-break-after: avoid; } h1 { font-size: 28px; border-bottom: 2px solid #213be7; padding-bottom: 0.3em; margin-top: 0; } h2 { font-size: 20px; border-bottom: 1px solid #c5c5d9; padding-bottom: 0.2em; } h3 { font-size: 16px; } p, li { color: #444656; margin-bottom: 1em; } ul, ol { margin-bottom: 1.5em; padding-left: 20px; } li { margin-bottom: 0.5em; } table { width: 100%; border-collapse: collapse; margin-bottom: 1.5em; page-break-inside: avoid; font-size: 13px; } th, td { border: 1px solid #c5c5d9; padding: 8px 12px; text-align: left; } th { background-color: #eeecfa; font-weight: 600; color: #1a1b24; } tr:nth-child(even) { background-color: #f4f2ff; } pre { background-color: #eeecfa; border: 1px solid #c5c5d9; border-radius: 8px; padding: 12px; overflow-x: auto; font-family: monospace; font-size: 13px; margin-bottom: 1.5em; page-break-inside: avoid; } code { font-family: monospace; background-color: #eeecfa; color: #1a1b24; padding: 2px 4px; border-radius: 4px; font-size: 13px; } pre code { background-color: transparent; padding: 0; border-radius: 0; color: inherit; } blockquote { border-left: 4px solid #973500; margin: 1.5em 0; color: #973500; background-color: #ffdbce; border-radius: 0 8px 8px 0; padding: 12px 16px; } blockquote p { margin: 0; color: #370e00; font-weight: 500; } /* Mermaid styles */ .mermaid { display: flex; justify-content: center; margin: 2em 0; page-break-inside: avoid; } /* Cover and header rules */ hr { border: 0; border-top: 1px solid #c5c5d9; margin: 2em 0; }

Daffodil Intelligence (DI) — Architectural Specification

High-Level Technical Design and Component Topology

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1. Executive Summary

Daffodil Intelligence (DI) is a 360° Multi-Agent Artificial Intelligence Operating System designed for academic institutions. DI centralizes academic, administrative, library, event, career, and research workflows into a single interface.

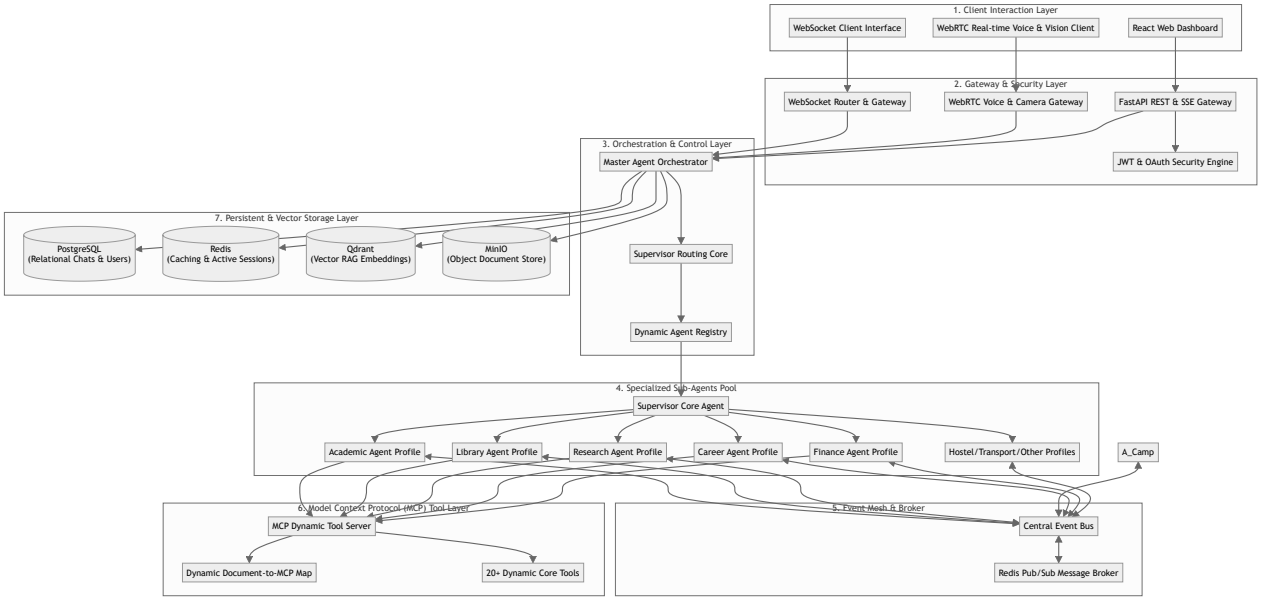
Rather than relying on static, single-agent architectures, Daffodil Intelligence employs a dynamic orchestrator model. This orchestrator dynamically compiles task goals, delegates processes across a mesh of specialized sub-agents, and interfaces with university databases and real-time communication channels (WebSockets, WebRTC voice endpoints).

This document outlines the system architecture, component topologies, and formal designs of the platform's core extensions: 1. **The Intercommunication Ecosystem (Agent-to-Agent Event Bus)** 2. **The Research Community & Publication Hub** 3. **The Dynamic Agent Spawning & Skill Hub System** 4. **The Comprehensive Dynamic Tool Catalog**

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2. System Architecture & Component Topology

Daffodil Intelligence utilizes a decoupled, multi-tier system architecture designed for high availability, secure data isolation, and real-time agent synchronization.



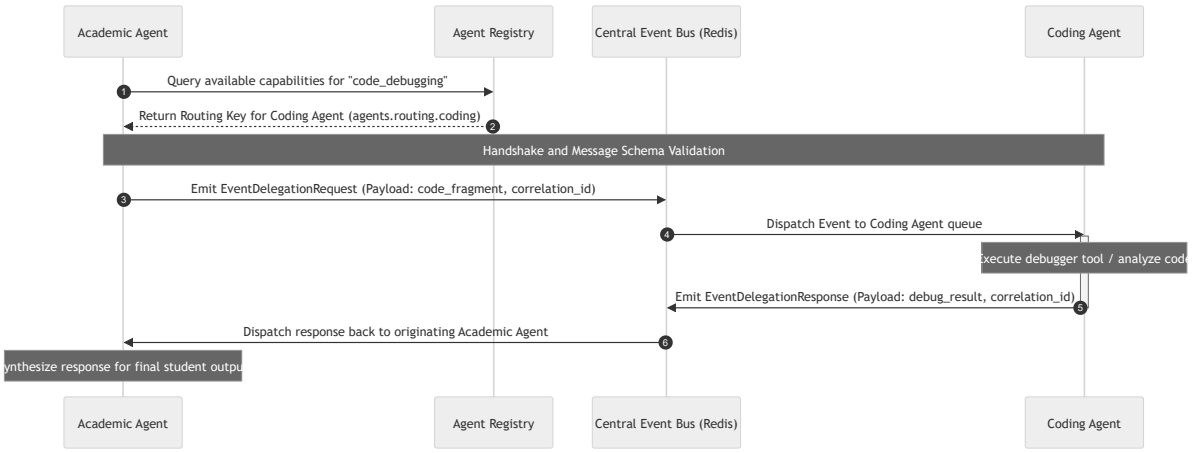
Component Tiers Description

1. **Client Interaction Layer**: Contains the visual workspace dashboard. Implemented with React and Zustand, it hosts the real-time text threads, dynamic agent metrics panels, and voice & live video communication channels (WebRTC).
2. **Gateway & Security Layer**: Built on FastAPI. Manages the entry gateways, executes authentication validations, decrypts JWT credentials, and routes connections to streaming pipelines (WebSockets, SSE).
3. **Orchestration & Control Layer**: The control hub of Daffodil Intelligence. It parses user inputs into task roadmaps, classification graphs, and dynamically resolves routing via the registry.
4. **Specialized Sub-Agents Pool**: Ready-made domain profiles acting as modular test cases. They represent target domain capabilities that can be activated on demand.
5. **Event Mesh & Broker**: Built on Redis Pub/Sub channels. Connects agents, allowing background, non-blocking task negotiation and result compilation.
6. **Model Context Protocol (MCP) Tool Layer**: Exposes standardized tools to agents, offering rate-limiting, validations, and dynamic document-mapping systems.
7. **Persistent & Vector Storage Layer**: Contains relational databases (PostgreSQL), high-speed memory caches (Redis), vector databases (Qdrant), and object repositories (MinIO).

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3. Core Subfeature: Intercommunication Ecosystem

The **Intercommunication Ecosystem** defines an asynchronous agent-to-agent (A2A) message-passing framework. This enables independent agents to negotiate, share insights, and delegate subtasks dynamically without manual user intervention.



3.1 Architectural Specifications

- **Central Event Bus**: Powered by a memory broker. Sub-queues are established dynamically for each sub-agent as they are spawned, routing inbound and outbound event messages.
- **Agent Registry**: Active agents advertise their state (`idle`, `thinking`, `executing`), tool capabilities, and communication keys.
- **Dynamic Event Routing**: When an agent determines a subtask exceeds its domain skills, it issues a delegation request to the event bus. The registry matches and routes the request to a qualified agent.
- **Agent-to-Agent Hub**: A centralized routing system managing direct message brokers, handling handshake protocols, and monitoring message queues to prevent timeouts.

3.2 Protocol & Schema Specifications

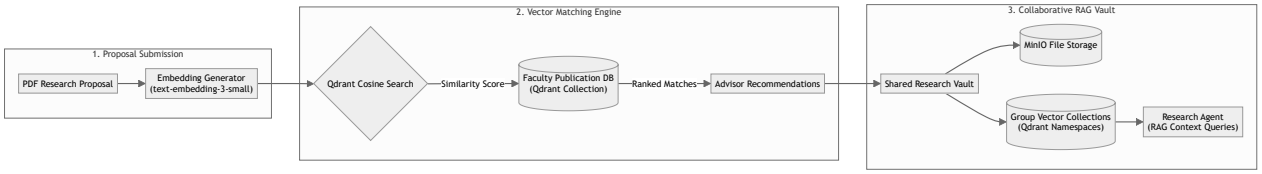
To ensure structural integrity across dynamic connections, all agent communication conforms to a strict, non-programmer structural specification:

Message Property	Type	Description	Example
Identifier (ID)	UUID	Unique identifier for the transaction event.	`f870d19e-f2ff-4ce9-94ea-9b888f04bb75`
Source	String	The originating agent system path.	`di.agents.academic`
Type	String	The transaction type (Request or Response).	`di.events.delegation.request`
Time	DateTime	ISO timestamp of event emission.	`2026-06-11T01:43:29Z`
Correlation ID	UUID	Session identifier mapping request-response pairs.	`a391ec96-edcc-3b9a-1135-c5c5d9e688f0`
Destination	String	The target agent receiving the event.	`di.agents.coding`
Data Payload	Object	Structured data containing the target task instructions.	`{ "required_action": "debug", "payload": { ... } }`

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4. Core Subfeature: Research Community Platform

The **Research Community Platform** bridges academic research, student initiatives, and thesis supervisions into a collaborative network.



4.1 Advisor Matchmaking System

- **Data Indexing**: Scrapes and indexes faculty research profiles, publication histories, and active project descriptions.
- **Semantic Mapping**: Student thesis abstracts or drafts are embedded and compared against the faculty registry in the vector database.
- **Matching Pipeline**: Generates ranked advisor recommendations based on cosine similarity scores, highlighting specific academic overlaps.

4.2 Shared Knowledge Vaults (RAG)

- **Joint Reading Vaults**: Research groups can create shared research folders. Uploaded PDFs are stored securely and partitioned by group.
- **Isolated Vector Indexing**: Document chunks are vectorized and isolated within group namespaces.
- **Collaborative Querying**: Members can query their shared vault via their research agent, ensuring RAG queries only retrieve documents belonging to their namespace.

4.3 Research Hub & Publication Feed

- **Reference-Link Ingestion**: Users showcase their research by adding reference links (e.g., DOI, arXiv, ResearchGate) instead of uploading the full text documents.
- **Publication Card Renderer**: Renders dynamic publication cards on user profiles and the global directory.
- **Trending Feed**: Background workers track views, citations, and semantic query matches to highlight trending papers.

4.4 Specialized Research Sub-Agents

- **AI Citation & Integrity Verification Agent**: Performs structural checks to detect generative AI fingerprints in drafts, flagging synthetic text patterns and verifying bibliography authenticity to produce an AI-similarity score.
- **Human-like Research Agent (Literature Review Specialist)**: Autonomously parses search databases, clusters papers, builds citation graphs, synthesizes alignments in literature, and drafts summaries.
- **Dynamic Ingestion to Agent Skill Memory**: When a user adds a research link, the document is vectorized in the background. The resulting embeddings are dynamically loaded into the student's personal `Research Agent` memory. This allows the student to converse with the publication using natural language.

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5. Dynamic Agent Spawning & Skill Hub System

Daffodil Intelligence features a dynamic execution layer allowing runtime modification of agent configurations, prompt behaviors, and capability bindings.

5.1 Programmatic Agent Spawning & Dynamic Sub-Agents

Rather than relying solely on the 18 pre-defined static agents, the `Master Agent Orchestrator` can programmatically spawn specialized sub-agents on demand: - **Runtime Compilation**: If a complex user goal is decomposed by the planning engine into highly specialized subtasks (e.g., analyzing a specific legacy COBOL file), the orchestrator dynamically creates an instance of a new sub-agent, compiles a target system prompt, binds requested tools, and registers it to the active session. - **Context Jailing**: Spawned sub-agents run in isolated execution blocks, inheriting only the necessary subset of conversation history and system parameters. - **Domain Profiles as Test Cases**: The 18 pre-configured agents function as modular test cases demonstrating this dynamic spawning engine.

5.2 Installable Agent Skill Hub (App Store Model)

The **Skill Hub** serves as a modular registry (analogous to a mobile app store) where custom tools, APIs, and computational libraries are organized as installable modules. - **App-Store Style Management**: The user interface provides a visual directory of all available skills (e.g., a "Bengali NLP Parser", "Financial Fee Calculator", or "VS Code Workspace Integration"). - **One-Click Binding**: When a user or supervisor agent clicks the "Install" button on a skill, the backend dynamically loads the skill's package and binds its functional entry point to the target agent. The agent instantly updates its system schema, declaring the new tool to the LLM during the next chat turn.

5.3 Own MCP Builder & Dynamic Document-to-MCP Server

To enable developers and agents to interface with external workflows, the platform provides: - **Own MCP Builder**: A developer utility that programmatically translates standard functions and API paths into a fully compliant Model Context Protocol (MCP) server container, generating dynamic schemas, validation rules, and endpoint bindings automatically. - **Dynamic Document-to-MCP Server**: Integrations that dynamically map uploaded files (PDFs, Word Documents, Excel sheets, text files) into a transient MCP server. Once a file is uploaded, the platform launches a dedicated MCP interface around it, exposing document-specific tools (e.g., `search_sections`, `retrieve_table_data`, `summarize_pages`). The target LLM interacts with the file directly via structured protocol tool-calls rather than consuming the raw, token-heavy text context.

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6. Multi-Channel Integration & Remote Orchestration

6.1 Multi-Channel Provider System (Adapter Architecture)

To provide unified access, Daffodil Intelligence abstracts external communication channels using a modular adapter gateway pattern. This functions similarly to automated node integrators (e.g., n8n adapter nodes): - **Provider Adapters**: Integration points ingest inbound messages and event triggers from Telegram, WhatsApp, and Google Account services. - **Unified Payload Mapping**: The gateway translates channel-specific payloads into the system's standardized message schema, passing the event to the orchestrator layer. - **Streaming Response Adapters**: Downstream token streams are transformed back into the format expected by the originating communication channel (such as Markdown for Telegram or templated payloads for WhatsApp).

6.2 CopyCat Remote Command & Control

Users can execute complex local machine workflows remotely through the multi-channel integrations: - **Remote Macro Invocation**: A authenticated user can send structured commands (e.g., via Telegram or WhatsApp) to trigger recorded CopyCat macro sessions. - **Virtual Session Execution**: The Master Agent resolves the command, retrieves the target macro script, and executes mouse and keyboard events on the host machine. - **Local Isolation Security**: To protect user privacy, the CopyCat system operates under a strict security boundary: - **No Cloud Exposure**: All captured inputs (mouse tracks, keyboard strokes, temporary coordinates) and UI screenshots remain strictly within the local loopback interface. - **Local Storage Encryption**: Execution logs and macro profiles are saved in encrypted folders on the host machine. - **Data Sanitization**: Prior to returning status summaries or logs to external LLMs or cloud endpoints, the system sanitizes text inputs to prevent data exfiltration.

6.3 Real-time Voice & Vision Integration

The real-time communication pipeline features dual media tracks for voice and live vision: - **Synchronized WebRTC Streams**: Handles concurrent audio and video feeds. The user's voice message is processed in parallel with their live camera stream or shared screen viewport. - **Visual Grounding Pipeline**: The vision system crops and parses the video frames to localize UI coordinates. The resulting annotations are combined with the audio transcript, providing the agent with real-time visual context.

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7. Dynamic Tool Catalog

The platform implements a comprehensive suite of domain-specific tools integrated into the Model Context Protocol (MCP) layer. These tools empower agents to read, query, compute, and interact with backing university services and data models.

7.1 Academic Domain Tools

- `get_cgpa`:
- *Function*: Retrieves the verified CGPA score of a student profile.
- *Details*: Performs secure queries against relational schemas to generate calculated semester averages.
- `get_courses`:
- *Function*: Fetches active course registrations for a specified semester.
- *Details*: Returns course titles, credits, syllabus links, and scheduled lecture timings.
- `get_results`:
- *Function*: Pulls official grade sheets and academic results.
- *Details*: Allows filter settings by semester and user ID, compiling scores dynamically.
- `check_prerequisites`:
- *Function*: Analyzes dependency chains for target courses.
- *Details*: Scans course directories to verify if a student has completed prerequisite courses.

7.2 Library Domain Tools

- `search_books`:
- *Function*: Queries the library database for physical and digital books.
- *Details*: Searches by query keywords, titles, or authors, outputting matches.
- `check_book_availability`:

- *Function*: Verifies circulation status and borrow history of books.
- *Details*: Details shelf numbers, copies available, and return timelines.
- `get_seat_availability`:
- *Function*: Monitors study study seat occupancy in reading rooms.
- *Details*: Emits real-time data feeds of vacant study zones.

7.3 Campus Services & Transport Tools

- `get_shuttle_schedule`:
- *Function*: Downloads shuttle routes and schedule changes.
- *Details*: Provides structured timelines organized by transit lines.
- `get_shuttle_location`:
- *Function*: Tracks transit positions along active routes.
- *Details*: Integrates with route maps to report real-time bus locations.
- `get_notices`:
- *Function*: Retrieves administrative bulletins and academic warnings.
- *Details*: Filters by department, category, or announcement date.

7.4 Career & Placement Tools

- `search_jobs`:
- *Function*: Searches job postings and internship listings.
- *Details*: Scans listings database by roles, salary grades, or corporate sponsors.
- `analyze_cv`:
- *Function*: Evaluates resume texts for compliance.
- *Details*: Runs semantic keyword assessments to highlight missing skills, formatting errors, and overall scores.
- `generate_career_roadmap`:
- *Function*: Generates developmental milestones for careers.
- *Details*: Produces step-by-step roadmaps, recommending certifications and projects based on career roles.

7.5 Financial & Tuition Tools

- `calculate_tuition`:
- *Function*: Calculates billing statements for active semesters.
- *Details*: Tabulates registered course credits, lab fees, and operational costs.
- `check_financial_dues`:
- *Function*: Audits transaction records for outstanding payments.
- *Details*: Reports payment details, waiver credits, and due dates.
- `get_scholarships`:
- *Function*: Details available scholarship programs and financial aid.
- *Details*: Returns application criteria, CGPA limits, and discount percentages.

7.6 Vision & Media Tools

- `analyze_screen`:
- *Function*: Runs deep image assessments on UI screen captures.
- *Details*: Identifies coordinate grids, text fields, buttons, and elements for cursor tracking.
- `extract_text_from_image`:
- *Function*: Performs high-speed Optical Character Recognition (OCR).
- *Details*: Translates document images or scans into plain textual blocks.

7.7 Semantic Memory Tools

- `search_memory`:
- *Function*: Scans session archives and vector records.
- *Details*: Queries semantic context history databases to find relevant references based on query inputs.

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8. Agent Ecosystem & Catalog

Daffodil Intelligence orchestrates 18 specialized domain agent profiles (which act as pre-configured test profiles):

#	Agent Class	System Identifier	Core Capabilities
1	**Supervisor Agent**	`supervisor`	Orchestration, intent classification, multi-agent dispatch, response synthesis.
2	**Academic Agent**	`academic`	CGPA calculation, course registration advice, syllabus reviews, semester GPA audits.
3	**Library Agent**	`library`	Book searches, electronic resources retrieval, physical book seat reservations.
4	**Campus Services Agent**	`campus`	Room booking management, campus announcements routing, facilities statuses.
5	**Career Advisor Agent**	`career`	Internship tracking, resume parsing & scoring, career roadmaps creation.
6	**Coding Agent**	`coding`	Code generation, syntactic debugging, technical interview training.
7	**Research Agent**	`research`	Literature discovery, citation styling, abstract reviews, thesis brainstorming.
8	**Wellness Agent**	`wellness`	Student stress management protocols, cognitive guidelines, academic study habits.
9	**Finance Agent**	`finance`	Fee structure auditing, scholarship waiver analysis, installment calendar tracking.
10	**Vision Agent**	`vision`	Document OCR, user interface screenshots parsing, charts/graphs interpretations.
11	**Voice Agent**	`voice`	WebRTC audio synthesis, voice message translation, transcription pipeline.
12	**Exam Agent**	`exam`	Midterm/Final schedules audit, seat planning locations, attendance tracking.
13	**Hostel Agent**	`hostel`	Residence hall checks, service requests, room vacancy management.
14	**Transport Agent**	`transport`	University shuttle tracking, route timelines, special event transport bookings.
15	**Event Agent**	`event`	Club activity coordination, workshop notifications, campus calendars.
16	**Complaint Agent**	`complaint`	Student grievance collection, ticket queue management, support escalation.
17	**Notification Agent**	`notification`	Multi-channel notifications routing (Telegram, email, WebSocket alerts).

#	Agent Class	System Identifier	Core Capabilities
18	**Fee Agent**	`fee_agent`	Tuition statement audit, billing disputes, waivers verification.